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Predicting Household Energy Burden

from Urban Form and Socioeconomic Characteristics in New York City

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04/30/2026

Goals of the Project

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Energy Burden in NYC

25% NYC households that qualify as energy burdened, defined as spending ≥ 6% income on energy costs

9.3% Median burden for low income households, vs. 2.9% citywide

$2.3B Excess energy costs across NYS low-income households (2022)

Impact & Equity

- High burden is linked to health deterioration, financial stress & housing instability (Hernández & Bird, 2010)

Urban Form Context

- No single dominant predictor: burden is shaped by localized combinations of environmental, social & economic conditions (Chun et al., 2025)
- Existing research focuses on consumption rather than burden, and in broader context. Burden must be understood on a granular level to capture localized conditions.

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To what extent do urban form characteristics improve the prediction of household energy burden beyond socioeconomic factors alone?

Policy Implications

Targeted Policy Interventions & Urban Planning

- Identifying strength and category of driving variables of energy burden (sociodemographic vs. urban form) improves ability to target policy to most impactful areas

Environmental Justice

- Quantifying the urban form in energy burden allows policies to address structural dimensions of inequity
- Income and economic measures can be complemented by place based investment

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Data and Methodology

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Data

Target Variable: Energy Burden (% of income spent on utility costs) - DOE LEAD Tool

Urban Form Variables

- Building age, lot size
- Floor Area Ratio (FAR),
- Bldg and residential area per unit, % residential
- Tree canopy coverage and temperature differential
- Subway stops (0.5 miles)
- Bus stops (0.25 miles)

Sociodemographic Variables

- Median Household Income
- % non-white population
- % renters
- Average household size
- % rent burdened households
- Number of households per tract

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Data Sources: DOE LEAD, PLUTO, U.S. Census ACS 5-Year Estimates, NYC Tree Census, NYC MTA Public Transportation Data, Climate and Economic Justice Screening Tool

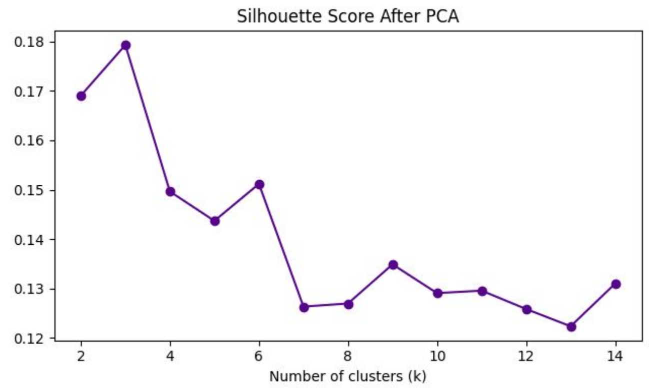
Methodology

Random Forest

- 80/20 Test Train Split, 5-fold cross-validation on training set
- RandomizedSearchCV: depth, n_estimators, min_samples_leaf, max_features
- Ablation
  - Linear Regression (socio only to establish baseline)
  - RF (socio only)
  - RF (all features)
- ΔR^2 isolates urban form contribution

PCA & K-Means Clustering

- PCA: 10 components retain 85% of variance
- K = 3 via silhouette (0.185) & elbow
- Hierarchical clustering discarded due degenerate splits (2,211 vs 1 obs.)
- K-Means chosen over GMM: hard cluster assignments support policy targeting
- Validation: Silhouette = 0.185, Calinski-Harabász = 424.5, Davies-Bouldin = 1.77



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Results

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Random Forest Ablation

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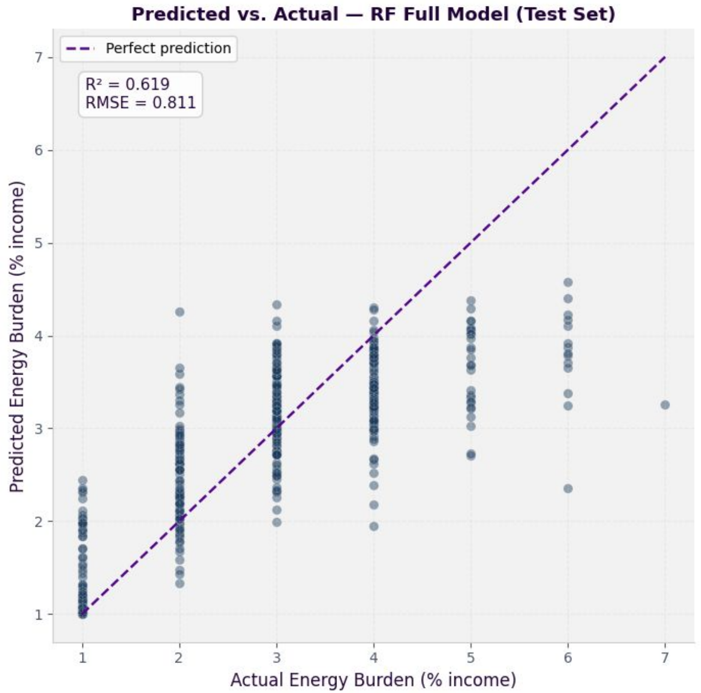
Model Performance



Model	RMSE	R^2
Linear Regression (socioeconomic only)	0.9251	0.4653
Random forest (socioeconomic only)	0.8723	0.5244
Random forest Tuned (socioeconomic only)	0.8677	0.5296
Random forest (Full)	0.8229	0.5760
Random forest Tuned (Full)	0.8183	0.5817

- Linear Regression baseline (socioeconomic only): $R^2 = 0.46$,
- RMSE = 0.92
- Random Forest (socioeconomic only): $R^2 = 0.52$, RMSE = 0.87
- Random Forest full model (all features): $R^2 = 0.58$, RMSE = 0.81
- $\Delta R^2 = +0.06$, Urban form explains an additional 6% of variance beyond socioeconomic factors alone

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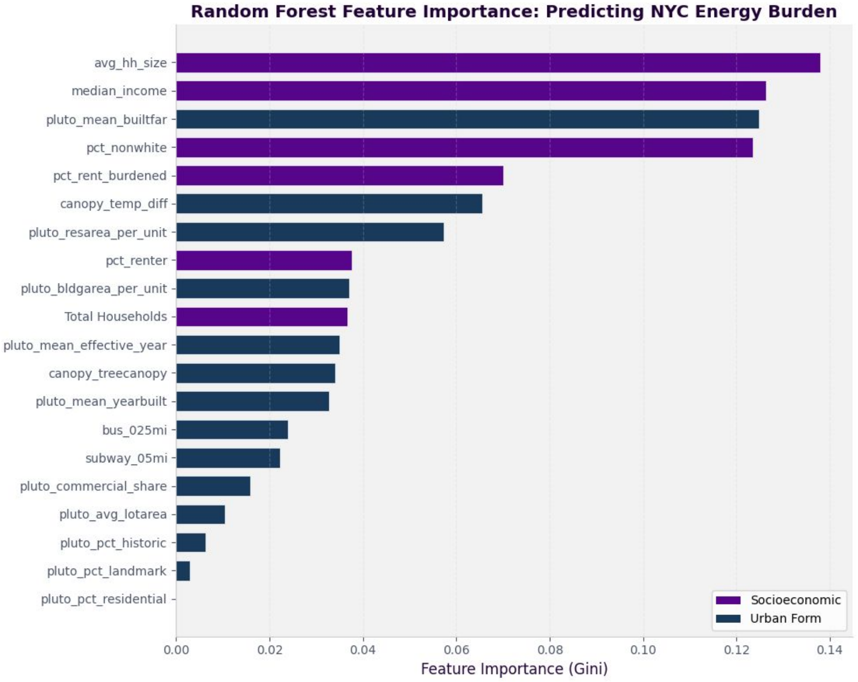
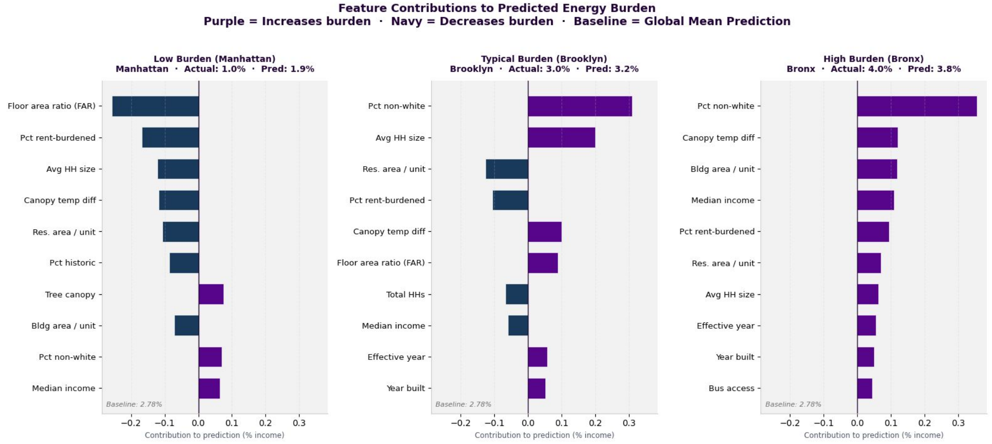
Feature Importance & Contribution





- FAR is the top urban form predictor, density lowers burden
- Urban form is protective in Manhattan tracts, absent in the Bronx
- Socioeconomic factors dominate, but urban form adds independent signal

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K-Means Clustering

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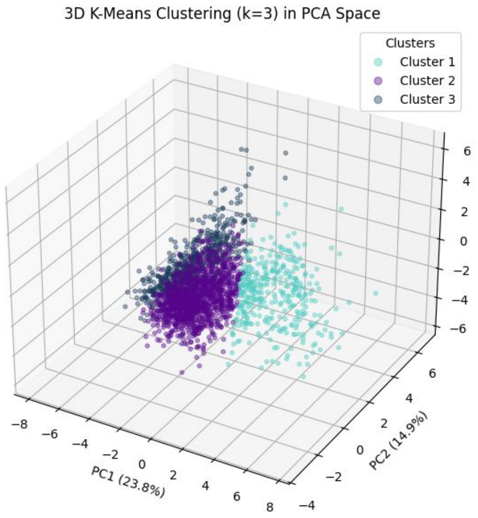
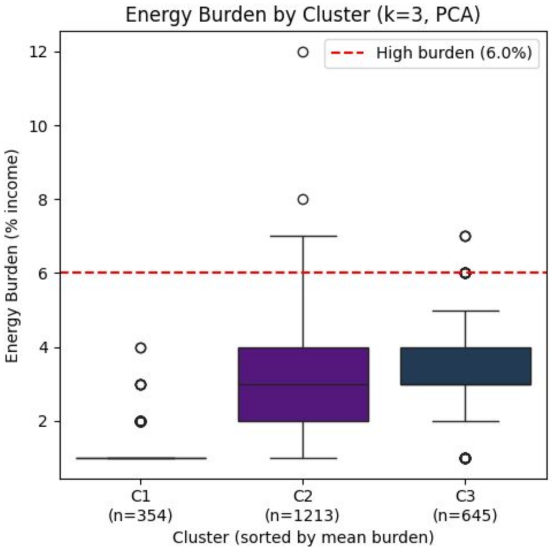
K-Means results





Cluster	1	2	3
Count	354	1213	645
Mean (Energy burden)	1.17 %	2.97 %	3.37 %
Standard deviation	0.45	1.2	0.97

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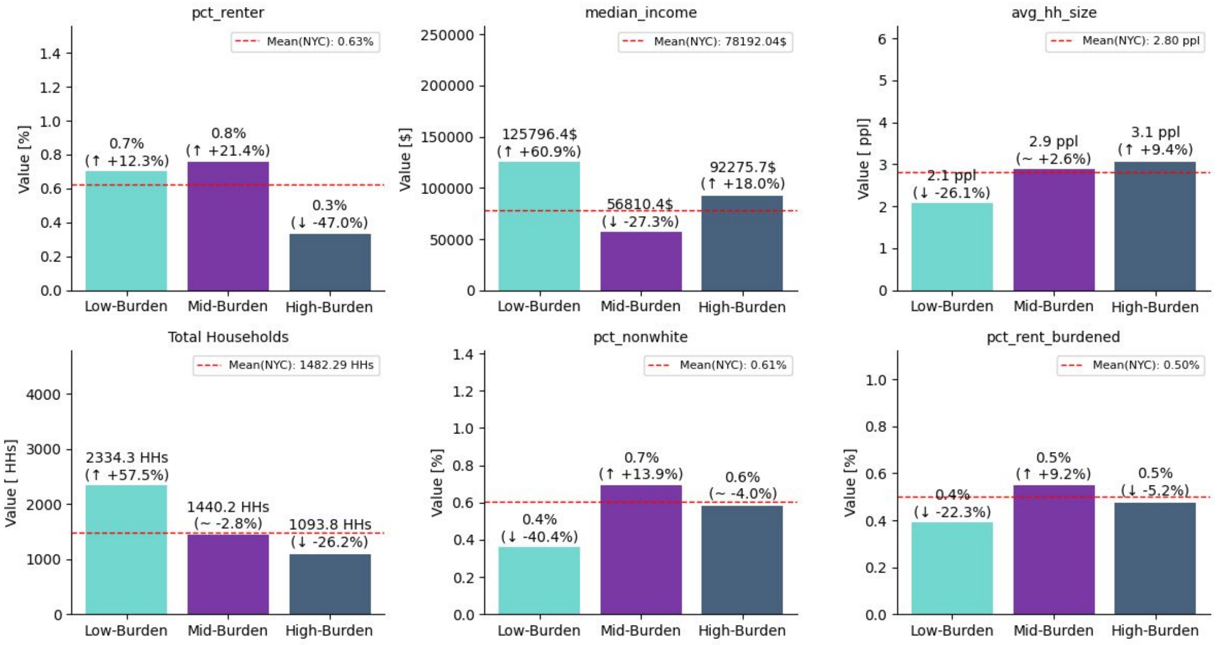
K-Means results

Sociological features

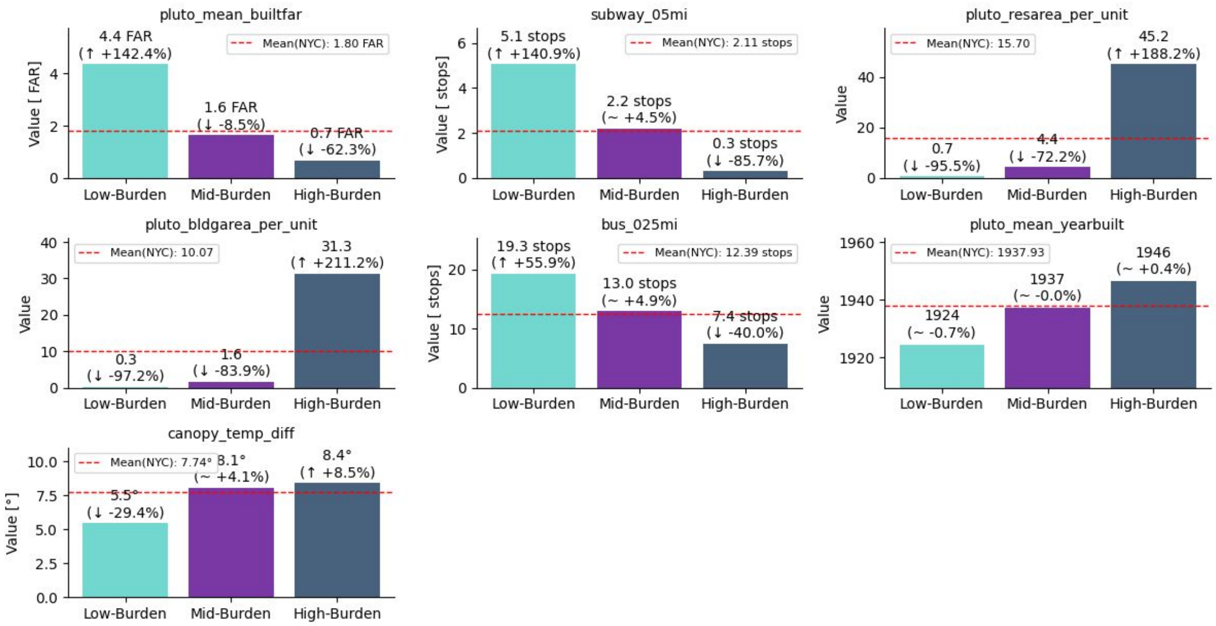


The Low-Burden cluster: higher median incomes, greater number of total households

The High-Burden cluster: lower percentages of renters, smaller total household counts

K-Means results

Non-Sociological (Urban form) features



The Low-Burden cluster: higher Floor Area Ratio (FAR), more transportation stops, older buildings, and lower temperature differences

The High-Burden cluster: lower FAR, fewer transportation stops, newer buildings, and larger area per units.

Discussion

Socioeconomic factors drive burden

- Median household income
- Percentage of non-white
- Percentage of being rent burdened

Urban Form Adds Explanatory Power

- Floor area ratio
- Proximity to transportation
- Air temperature differences
- Building age

Limitations

- Tract-level resolution limits household targeting.
- LEAD estimates are modeled, not metered.
- Staten Island results should be interpreted with caution due to lower population density.

Recommendations

- Increase awareness of existing programs
  - Low Income Home Energy Assistance Program
- Make existing policies stronger and apply to all buildings
  - Local Laws 84 and 97
- **Short-term**: expanded energy vouchers and encourage demand response enrollment
- **Long-term**: retrofitting buildings → reduce structural demand; increasing green space → mitigate urban heat island effects

Conclusion

Energy burden is a highly **interconnected issue within urban systems**, impacted by poverty, environmental justice, and accessibility.

Energy burden in NYC is **driven primarily by socioeconomic disadvantage**, but urban form plays a measurable role, particularly where poor building conditions and limited transit access compound economic vulnerability. **Addressing it requires both targeted financial relief and long-term investment in the built environment**.

Sources

Chun, J., Ortiz, D., Jin, B., Kulkarni, N., Hart, S., & Knox-Hayes, J. (2025). Energy burden in the United States: An analysis using decision trees. Energies, 18(3), 646. https://doi.org/10.3390/en1803064

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